Week 2 HW

1. shape 8x8x1, kernel size 3x3

$$H_{out} = \left[\frac{H+2P-K}{S}\right]+1$$

$$= \left[\frac{8+0-3}{1}\right]+1$$

$$= 6$$

The output shape is 6x6x4

2. $$H_{out} = \left[\frac{H+2P-K}{S}\right]+1$$

$$= \left[\frac{10+2\cdot 1-3}{1}\right]+1$$

$$= 10$$

The output shape is 10x10x2

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3. $3 \times 3 \times 3 \times 1 = 27$ weights

$27 + 1 \times 3 = 30$ parameters

4.

$$\text{Output} = \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$$

5. $$\text{Output} = \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$$

6. A filter in a CNN is a small matrix of learnable features, and it detects features such as vertical and horizontal edges in the graph.

7. Padding adds additional borders of pixels on the edge input.

We add padding to original data so we can control the dimension of our output accordingly.

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8. Max-pooling chooses the largest value within ~~the~~ a matrix and downsamples the original matrix to its largest value.

We use max-pooling b.c. we want to downsize the input, while keeping the sharpest charachteristics of the input.

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